demonstrating how aerial vantage points could be used for surveillance and intelligence gathering.

Observation balloons were first employed during the French Revolutionary Wars (1794) when the French army used the “l’Entreprenant” hydrogen balloon to observe enemy troop movements at the Battle of Fleurus. Throughout the American Civil War (1861–1865) and later conflicts, tethered balloons became standard tools for battlefield reconnaissance, allowing armies to map terrain, direct artillery fire, and monitor enemy positions from above. These balloons were often manned but laid the conceptual groundwork for remote aerial observation, a defining function of later drones.

Kites also played a significant role in the early development of aerial imaging and meteorological research. In the late 19th century, Arthur Batut (1888) and Lawrence Hargrave (1890s) experimented with kite aerial photography (KAP) attaching cameras to kites to capture images from the sky. These unmanned photographic experiments represented some of the earliest attempts to gather data from the air without direct human presence onboard. By the early 1900s, meteorologists used kite systems to collect atmospheric data, temperature, and pressure at various altitudes establishing the principle of remote sensing, which would later become integral to UAV-based environmental monitoring.

2.4 Technological Foundations

The technological foundations of first-generation drones were established during the early 20th century when engineers and military scientists began experimenting with remotely piloted aircraft. Although primitive compared to today’s UAVs, these early systems introduced the core engineering principles that shaped the evolution of unmanned flight namely, remote